

# Q1 2024 Earnings Call

## Company Participants

- Brice Hill, Senior Vice President & Chief Financial Officer
- Gary E. Dickerson, President and Chief Executive Officer
- Michael Sullivan, Corporate Vice President

## Other Participants

- Atif Malik, Analyst, Citi
- Brian Chin, Analyst, Stifel
- CJ Muse, Analyst, Cantor Fitzgerald
- Charles Shi, Analyst, Needham & Company
- Christopher Caso, Analyst, Wolfe Research
- Harlan Sur, Analyst, J.P. Morgan
- Joseph Moore, Analyst, Morgan Stanley
- Joseph Quatrochi, Analyst, Wells Fargo
- Krish Sankar, Analyst, TD Cowen
- Srinii Pajjuri, Analyst, Raymond James
- Stacy Rasgon, Analyst, Bernstein Research
- Thomas O'Malley, Analyst, Barclays
- Timothy Arcuri, Analyst, UBS
- Toshiya Hari, Analyst, Goldman Sachs
- Vivek Arya, Analyst, Bank of America

## Presentation

### Operator

Welcome to the Applied Materials' Earnings Conference Call. During the presentation, all participants will be in a listen-only mode. Afterwards, you will be invited to participate in a question-and-answer session.

I would now like to turn the conference over to Michael Sullivan, Corporate Vice President. Please go ahead, sir.

### Michael Sullivan {**BIO 16341622 <GO>**}

Good afternoon, everyone, and thank you for joining Applied's First Quarter of Fiscal 2024 Earnings Call. Joining me are Gary Dickerson, our President and CEO; and Brice Hill, our Chief Financial Officer.

Before we begin, I'd like to remind you that today's call contains forward-looking statements which are subject to risks and uncertainties that could cause our actual results to differ. Information concerning the risks and uncertainties is contained in Applied's most recent Form 10-K filing with the SEC.

Today's call also includes non-GAAP financial measures. Reconciliations to GAAP measures are found in today's earnings press release and in our quarterly earnings materials which are available on our website at [ir.appliedmaterials.com](http://ir.appliedmaterials.com).

Before we begin, I have a calendar announcement. On Monday evening, February 26th, Applied will host a panel at the SPIE Advanced Lithography and Patterning Conference in San Jose. Joining us will be leading experts from NVIDIA, Intel, IMEC, and Siemens EDA. We will also have demo stations with several new products and technologies, will be introducing at the event. There won't be a webcast so we hope you'll join us in San Jose.

And with that introduction, I'd like to turn the call over to Gary Dickerson.

## **Gary E. Dickerson** {**BIO 2135669 <GO>**}

Thank you, Mike. Applied Materials made a strong start to fiscal 2024 with first-quarter revenue in the high end of our guidance and earnings that exceeded our guided range. Our inflection-focused innovation strategy is delivering results. We have outperformed our markets for five consecutive years and believe we are in a great position as customers transition major new chip innovations to high-volume production over the next several years.

The breadth of our technology capabilities combined with our deep customer relationships, allows us to see inflections early and accelerate key technology innovations that are critical to scaling AI, IoT, electric vehicles, and renewable energy. We have reshaped and expanded our portfolio of solutions that enable next-generation transistors, new interconnect schemes including backside power delivery, high-performance DRAM including high-bandwidth memory, and specialty applications in the ICAPS market.

In my prepared remarks today, I'll provide some examples of how these inflections grow Applied's available market and are highly accretive to our share. I'll also talk about our long-term strategy to accelerate innovation and commercialization velocity through tighter collaboration with our customers and partners. But to begin, let me share our latest perspective on the market environment.

In our discussions with customers, we're hearing that overall market dynamics are improving. There is a re-acceleration of capital investment by cloud companies, fab utilization is increasing across all device types. And memory inventory levels are normalizing. In terms of Applied's business in 2024, we see leading-edge foundry logic being stronger year-over-year, even though some important projects are delayed. We're forecasting ICAPS demand to be slightly lower than 2023 with weakness in some end-markets being offset by strong regional investments.

We expect our NAND revenues to be up year-on-year but NAND to remain less than 10% of total wafer fab equipment spending. And we see continued strength in our DRAM business, driven by customers ramping production of high-bandwidth memory. High-bandwidth memory or high-performance DRAM dies are stacked and connected to logic die with advanced packaging, is a key enabler for the AI data center.

The dies used in high-bandwidth memory are more than two times larger than standard DRAM, which means that more than twice the capacity is needed to produce the same volume of chips. On top of this the packaging steps needed for die stacking further increase our total available market. High-bandwidth memory or HBM made up only about 5% of DRAM output in 2023 but is expected to grow at a 50% compound annual growth rate over the coming years.

DRAM is a great example of how our inflection-focused innovation approach is working. By focusing on the critically enabling process and packaging steps for next-generation technologies, Applied has significantly increased our share of the DRAM market. In 2023, we estimate that our DRAM share was more than 10 points higher than it was a decade earlier. And our DRAM revenues were larger than our two closest process equipment peers combined.

We're also best positioned for future growth. Thanks to our leadership and logic technologies that have been implemented for DRAM peripheral circuitry applications to enable significantly increased I/O speed. Our strong position in DRAM patterning, our unique co-optimized hard mask solutions, which are critical for capacitor scaling. And advanced packaging, where we have strong leadership positions and micro-bump and Through Silicon Via that will enable multiple generations of high-bandwidth memory.

In fiscal 2024, we expect our HBM packaging revenues to be four times larger than last year, growing to almost \$0.5 billion. And across all device types we expect revenue from our advanced packaging product portfolio to grow to approximately \$1.5 billion. Looking further ahead, we see opportunities for this business to double again as heterogeneous integration is more widely adopted and we introduce new products that expand our served market.

Another key inflection that will transition to high-volume production beginning this year is Gate-All-Around transistors in leading-edge foundry logic. These complex 3D structures can provide a more than 30% improvement in a chip's energy efficiency. This is especially enabling for high-performance AI data center applications. The shift from FinFET to Gate-All-Around grows Applied's available market by \$1 billion for every 100,000 wafer starts per month of capacity. And we're on track to gain share and capture over 50% of the spending for the process equipment used in this new transistor module.

Major advances in leading-edge foundry, logic, and DRAM are also driving the need for more and better metrology and inspection to be integrated into the manufacturing flow. We have developed industry-leading cold field emission eBeam Technology that enables highly sensitive 2D and 3D imaging at up to 10 times higher speeds. We expect our CFE systems revenue to grow by a factor of four in 2024 and represent 50% of our total eBeam system sales.

The incredible innovation we see in the industry today is not limited to the leading edge. In recent years ICAPS customers have invested about 10% of their revenues or about \$30 billion annually in research and development to accelerate the roadmap for IoT, communications, automotive, power, and sensor technologies. ICAPS technology depends less on shrinking device features and customer investments are heavily weighted towards new structures, new materials, and new integration approaches playing to the core strengths of Applied.

ICAPS is another area where we saw market inflections early. And five years ago, we formed a dedicated team to focus on the needs of these customers. Since then, we've released more than 20 new ICAPS products that target the highest-value device innovations in these markets. And we have a robust development pipeline of unit process and integrated solutions. While major end-market inflections such as AI and IoT, Electric vehicles, and renewable energy are already driving semiconductor growth and innovation, it's important to recognize they are still in the early stages of adoption.

For example, high-performance GPUs for AI data centers only represent 6% of leading-edge foundry logic wafer starts today. The full potential of technologies like AI cannot be unlocked without next-generation chips with better performance, power, and cost. The technology roadmap for semiconductors is rich with possibilities and opportunities, but also incredibly complex. No company is better placed to address this complexity than Applied Materials. With the industry's broadest and deepest portfolio of capabilities and products, we have a unique ability to combine, co-optimize, and integrate our technologies to develop highly differentiated solutions for our customers.

To bring these advances to market faster, we're also innovating the way we innovate, by driving earlier and deeper collaboration with our customers and partners. We are expanding our global innovation network that will connect into the EPIC Center we're building in Silicon Valley. During the quarter, we announced an expansion of our long-term partnership with LETI which is focused on accelerating ICAPS innovation, and we launched a new collaboration with MIT which is centered around next-generation power electronics. As industry complexity rises, we are also delivering more value to customers with our advanced services that enable our customers to accelerate R&D, transfer new technology into volume manufacturing faster, and then optimize yield, output, and cost in their factories.

AGS has delivered 18 consecutive quarters of year-on-year growth. Revenue for the first quarter was up 8% versus the same period last year and the business is now at a \$6 billion annual run rate. AGS has the opportunity for double-digit growth this year, and we believe we can sustain this growth rate into the future. A significant portion of AGS revenue is generated from subscriptions. We have almost 17,000 tools under service agreements up 8% year-on-year and these agreements have a very high renewal rate over 90%.

Before I pass the call over to Brice, I will quickly summarize. Applied Materials outperformed our markets in 2023 for the fifth consecutive year and we delivered strong results in the first quarter of 2024. The positions we've established at key industry inflections will support continued outperformance as customers ramp next-generation chip technologies into high-volume production. We are strengthening R&D collaboration with customers and partners to drive innovation and commercialization velocity, improvements in mutual success rate, and R&D investment efficiencies. And we see growing demand for our advanced services that are helping customers manage increasing complexity in their business as the industry scales.

Now, I'll hand over to Brice.

## **Brice Hill** {**BIO 19639921 <GO>**}

Thank you, Gary. And I'd like to thank our teams for delivering strong revenue and margins this quarter and making further improvements in our operating performance. On today's call, I'll discuss our value creation strategy and the results it is producing. Then, I'll summarize our growth thesis and why we believe we will outperform our markets in the years ahead. Finally, I'll summarize our Q1 results and provide our guidance for Q2.

I'll begin by discussing how our assets and strategy create value for shareholders. Applied has the broadest and deepest process equipment portfolio and expertise in the industry. We are highly invested in collaborating with our customers, allocating \$3 billion

in annual R&D to invent new solutions to the most critical semiconductor manufacturing challenges. Increasingly, the only way to solve these challenges is by co-optimizing and integrating our chamber technologies in new ways. In addition, identifying new materials and processes early and collaborating closely with customers leads to faster results, a higher probability of success, greater efficiency, and stronger financial returns.

The benefits of our value creation strategy are being demonstrated in our financial results. We generated record equipment sales of \$20.7 billion in calendar '23, including legacy equipment reported in AGS, and we extended our strong position in DRAM with record calendar year sales of over \$4.3 billion. In fact, over the past 10 years, the company has gained over 10 points of DRAM share and multiple points of overall share. This has contributed to Applied delivering a fifth straight year of overall WFE share gains and one of the best share outcomes of the past 20 years.

Over the same 10 fiscal years, we've grown company revenue at a compound rate of over 13%, non-GAAP EPS at nearly 30%, free cash flow at 33%, and dividends per share at nearly 12%. Also over this period, we increased return on invested capital from 8% to 35% and reduced net shares outstanding by over 30%.

Next, I'll summarize our growth thesis. As we look out over the planning horizon, we expect semiconductors to grow significantly faster than GDP. Second, we expect the equipment market to grow as fast or faster than semiconductors over time, driven by increasing technical complexity. Third, we expect Applied's equipment business to outgrow the market. And fourth, we expect our services business to grow as fast or faster than our equipment business.

I'll take a moment to support the third pillar of our thesis. That Applied's equipment business will outgrow the market. The reason is that our technologies enable the key semiconductor advances needed to drive growth in AI, IoT, and renewable energy.

Looking ahead to the semiconductor process inflections that will play out over the next several years, the company is extremely well positioned. In data center AI, we are number one in process equipment for advanced logic and compute memory, both standard DRAM and high-bandwidth memory. We also have line of sight to share 50% or more in gate-all-around transistors, backside power delivery, and advanced packaging.

We are equally strong in edge AI and IoT, with the number one position in ICAPS silicon which is used to sense and convert analog information and transmit it to the cloud. We are also innovating rapidly in ICAPS technology for the global energy transformation, including through new agreements with partners like LETI and MIT, which Gary described. In summary, we feel confident that our unique assets and collaboration strategy position Applied to continue to outpace our markets and deliver strong shareholder returns as these major inflections play out over the next several years.

Now I'll summarize our Q1 results. On a year-over-year basis, net sales declined slightly to \$6.7 billion. Non-GAAP gross margin grew 110 basis points to 47.9%. Non-GAAP OpEx grew 5.6% to \$1.23 billion and non-GAAP EPS grew nearly 5% to \$2.13.

Turning to our segment results, Semiconductor Systems revenue was strong at \$4.91 billion and included record DRAM and etch system sales. Segment non-GAAP operating margin was 35.7%. While our operating expenses are primarily focused on R&D programs for emerging technology inflections, we are also investing to expand and diversify our manufacturing, logistics, and supply chain to efficiently serve future growth.

Applied Global Services delivered record revenue and its 18th consecutive quarter of year-over-year growth. AGS revenue increased approximately 8% year-over-year to nearly \$1.48 billion, and segment non-GAAP operating margin was 28.3%. Our installed base surpassed 49,000 tools during the quarter and grew to nearly 200,000 chambers. Around two-thirds of AGS recurring services and parts revenue was delivered as subscription agreements. Finally, AGS continued to produce more than enough operating profit to fund Applied's growing dividend.

Moving to Display, Q1 revenue was \$244 million, and segment non-GAAP operating profit was 10.2%. We continue to look forward to our opportunity in the upcoming OLED IT growth inflection.

Turning to cash flows in Q1, we generated \$2.3 billion in operating cash flow and \$2.1 billion in free cash flow. We distributed \$966 million to shareholders, including \$266 million in dividends and \$700 million in buybacks. We repurchased nearly 5 million shares at an average price of \$152.60.

Please note that our Q1 results include the following. First, as we discussed in our recent 10-K report, we increased the estimated useful lives of our plant and equipment, and this increased non-GAAP EPS by \$0.03. Also effective Q1, we refined the way we allocate stock-based compensation, moving the majority of the expenses from corporate unallocated to the operating segments, which gives managers greater visibility over costs. While the change has no impact on company operating profit or EPS, it reduces segment operating profit and corporate unallocated costs proportionally.

To help you with your segment models, our quarterly earnings presentation includes a table showing what operating profits would have been in fiscal 2022 and in each quarter of fiscal 2023 on a like basis. Finally, the reduction in depreciation and share-based compensation in cost of sales increased gross margin by approximately 40 basis points.

Now, I'll share our guidance for Q2. We expect revenue to be \$6.5 billion, plus or minus \$400 million. And we expect non-GAAP EPS of \$1.97 plus or minus \$0.18. Within this outlook, we expect Semi Systems revenue of around \$4.8 billion, AGS revenue of about \$1.5 billion, and Display revenue of around \$150 million. We expect non-GAAP gross margin to be approximately 47.3% and non-GAAP operating expenses to be around \$1.235 billion. We are modeling a tax rate of 12.5%.

Thank you, and now Mike, let's begin the Q&A.

**Michael Sullivan** {[BIO](#) [16341622](#) <GO>}

Thanks, Brice. And our goal is to help as many of our analysts as possible. With that in mind, please ask just one question on today's call. If you have another question, please re-queue and we'll do our best to come back to you later in the session.

Operator, let's please begin.

## Questions And Answers

### Operator

Certainly. (Operator Instructions) Our first question comes from the line of Stacy Rasgon from Bernstein Research. Your question, please.

**Q - Stacy Rasgon {BIO 16423886 <GO>}**

Hi guys, thanks for taking my question. I wanted to ask about DRAM in China. So DRAM was very strong and was supposed to be strong, but it was quite a bit stronger than I think we thought it was going to be. I guess, can you tell us how much of that was China versus non-China and going forward last quarter you talked about the expectations for the China piece of that had been pulled forward due to the sanctions to roll off as we went through the rest of the year. What are your thoughts on that China trajectory as we go into April quarter and into the second half?

**A - Brice Hill {BIO 19639921 <GO>}**

Hi, Stacy. Thanks for your question. So in the current -- in the quarter we just closed, we did see high shipments of China DRAM, and we -- it was approximately the same in terms of the higher quantity as we saw in Q4. And just to be clear, we'll expect another quarter in Q2, in our outlook quarter that it should remain elevated. I think for the Q4, we had said it was approximately \$500 million increase on the DRAM side. That's probably a good estimate for all of those quarters.

And then the second part of the question. As we look through the rest of the year, we'll expect that to normalize -- our China mix should normalize from the levels it's out right now to something that's more typical with our average.

**Q - Stacy Rasgon {BIO 16423886 <GO>}**

Which is what -- is that -- what's the typical?

**A - Brice Hill {BIO 19639921 <GO>}**

Well, I would say from a long-term -- from a many year perspective, we average approximately 30%, so if we're at 45% right now, will decline across the year to somewhere around that level.

**Q - Stacy Rasgon {BIO 16423886 <GO>}**

Got it. That's helpful guys. Thank you.

**A - Gary E. Dickerson {BIO 2135669 <GO>}**

Thank you.

**Operator**

Thank you. One moment for our next question. And our next question comes from the line of Vivek Arya from Bank of America. Your question please.

**Q - Vivek Arya {BIO 6781604 <GO>}**

Thank you for taking my question. I was hoping if you could give us your view on how you see the WFE environment in '24. I think many of your competitors have suggested kind of a low- to mid-single-digit growth. But within that, the move towards more leading edge and DRAM and less on the trailing edge. But just given how strongly Applied grew in some of your trailing edge and ICAPS, does -- how does that position you in this new WFE environment? Because I think I know you said that you only see only limited decline on



the ICAP (Sic - ICAPS) side, but when I look at the CapEx of many of the auto industrial and analog companies in the U.S., they are cutting it quite sharply. So that's why I was curious why you think that it's only going to decline. So just broadly comment on WFE and the different piece parts. And then maybe a China versus non-China kind of cutting of that. Thank you.

**A - Brice Hill {[BIO 19639921 <GO>](#)}**

Okay. Thanks, Vivek. No real change I think in our outlook for '24 in terms of shaping how those end-markets are evolving. So we do think DRAM will continue to be a strong market. We think NAND will improve from its low levels a little bit. We think leading logic will be larger as Gate-All-Around and new investments start to ramp towards the back half of the year. And we do think there will be some digestion in ICAPS and China, both. Just to be crystal clear, we had enormous growth for two years in ICAPS and the China-related ICAPS business. And so we won't see that enormous growth this year, it may be a little bit smaller. We think there is some digestion with that capacity, but we expect that market to grow over time along with the underlying rates for the company. So that's -- those -- the shape of those end-markets hasn't changed in our outlook.

**Q - Vivek Arya {[BIO 6781604 <GO>](#)}**

Is that consistent with the low- to mid-single-digit WFE that others are suggesting or do you have a different view?

**A - Brice Hill {[BIO 19639921 <GO>](#)}**

Well, all we can do is tell you what we see from Applied's perspective. When we commented on '23, we said it was a strong year for Applied. And to the way you asked your question, we had strong ICAPS, we had strong DRAM, we were strong in growth and packaging. And so as we look towards '24, those are the puts and takes we're not going to give a precise number for '24.

**Q - Vivek Arya {[BIO 6781604 <GO>](#)}**

Thank you, Brice.

**Operator**

Thank you. One moment for our next question. And our next question comes from the line of CJ Muse from Cantor Fitzgerald. Your question, please.

**Q - CJ Muse {[BIO 6507553 <GO>](#)}**

Yeah, good afternoon, thanks for taking the question. I guess, given the strong outperformance that you showed in '23 where your silicon business actually grew, where I think most people are thinking WFE down in the high-single-digits. Curious how you're thinking about 2024. You've talked about, clearly, you're benefiting from share gains across leading-edge foundry, logic, and DRAM. But also vision for ICAPS slowing. So do you think it's another year of outperformance or is it a year of digestion? If you could kind of walk-through that. And just as a kind of a bonus question, with SPIE just a little over a week away, would you care to give a preview of what we will hear including a focus on Sculpta? Thanks so much.



## A - Brice Hill {BIO 19639921 <GO>}

Okay, so I'll tackle the first part and then I know Gary wants to tackle the second part of the question. So on outperforming in '24, CJ, we do expect, because of our exposure to the fast-growing markets and some of the inflections growing quickly in '24. We do expect to outperform. We're not making a call on the size of the market. It's like you said, there is a couple of markets that are growing, the ICAPS market and the China piece, we think won't grow. And so we're not making a call on which is the stronger trend and I don't know if we know. So we'll see how the year plays out from that perspective.

And Gary, on the show.

## A - Gary E. Dickerson {BIO 2135669 <GO>}

Hi, CJ. On -- just let me start on the outperformance. Yeah, I think the most important thing to think about is how we're positioned for major inflections. So if you think about foundry logic leading-edge, Gate-All-Around, backside power distribution, those are incremental billion-dollar opportunities for Applied, where we have an opportunity for more than 50% share, which is very accretive to our overall market share. So we're really well-positioned there. ICAPS, we formed that group five years ago, as I said earlier, 20 major new products have been introduced. We have a strong pipeline of future ICAPS products. And so, again there -- and we have opportunities to grow in segments like etch and PDC, where we have a lot of momentum. So I like our position in ICAPS. DRAM, we've gained more than 10 points of share over the last 10 years.

And as I mentioned in the prepared remarks, extremely well-positioned for the major inflections in DRAM. In packaging, we have the strongest and broadest portfolio. And this is around \$1.5 billion of revenue for us in '24 and an opportunity to double over the next few years. So all of those areas, I think really set us up for continued outperformance.

And then on your question about SPIE. One of the things we'll be talking about there is Sculpta, just reminding people that's a breakthrough pattern shaping technology, that provides a simpler, faster, and more cost-effective alternative to EUV double patterning. So we're engaged with all of the leading foundry logic customers and expanding Sculpta steps for advanced patterning, including High-NA EUV. And we're also working with customers on new Sculpta applications. And we expect this business to grow to close to \$200 million in 2024 and ramp to around \$0.5 billion in annual revenue in the next few years.

Also at SPIE, for those of you that who will attend, you'll hear about new etch and CVD technology for patterning. That will be very large growth drivers for the company and enable us to continue to outperform. And just for reference in patterning, we've increased our served market from around 1.5 billion, 10 years ago, to 8 billion now. And our share from around 10% to 30%. So when Brice said I was excited, I'm absolutely excited. These are some really, really great technologies with very strong customer pull and delivering meaningful growth for the company.

## Operator

Thank you. One moment for our next question. And our next question comes from the line of Chris Caso from Wolfe Research. Your question, please.

## Q - Christopher Caso {BIO 4815032 <GO>}

Yes. Thank you. Good afternoon. I guess the question is kind of looking at the order rates, and more importantly, what your customers are telling you as you're looking into calendar '25. As you know, some others in the industry with long lead times have started to see some of those green shoots coming into '25. You know that, we're balancing here between some of your customers burning off capacity and going through technology transitions. What's the thought, as we start to look into '25, at these early days?

**A - Brice Hill {BIO 19639921 <GO>}**

Hi, Chris. Thanks for the question. When we look at the market currently, we're seeing improvements in inventories and we're seeing improvements in utilization. So it's starting to pick up, that's pretty much across -- that is across the entire market on the utilization side. And then what we're hearing from customers is optimism, generally speaking for 2025. We would echo comments we've heard from others that the semiconductor end-market for devices is expected to be growing and its investment cycle on the leading edge, we're expecting the memory markets to continue to improve. So '25, we are optimistic about the direction for '25.

**Q - Christopher Caso {BIO 4815032 <GO>}**

Thank you.

**Operator**

Thank you. One moment for our next question. And our next question comes from the line of Krish Sankar from TD Cowen. Your question, please.

**Q - Krish Sankar {BIO 16151788 <GO>}**

Yeah. Hi. Thanks for taking my question. Gary, I had a question for you. You have a broad-based product portfolio, you outperformed WFE last two years. I'm kind of curious for some of these new applications whether it's HBM, Gate-All-Around, or even backside power delivery. Our customers looking at a one-stop shop or more going with best-of-breed solution. In other words, let's say, for Gate-All-Around, it's the strength in Epi, helping your Etch or ALD products.

**A - Gary E. Dickerson {BIO 2135669 <GO>}**

Yeah. Chris, one thing I would say, that for all of these inflections, there is a tremendous advantage for us to have that broad portfolio. Again, when you think about one of the examples, I've used many times is your processor chip and your smartphone, with 15 billion transistors, and 60 miles of wiring, which is kind of mind-boggling, when you think about how do we do that, how do we create something like that.

There are over 1,500 steps in building that type of chip. And when you're developing these new technologies like Gate-All-Around, like backside power or DRAM technologies, or any of these packaging technologies. The ability to combine and co-optimize these steps is an enormous advantage. Enormous -- that 60 miles of wiring, we have one platform that combined seven technologies under vacuum. To enable those 60 miles long very thin wires to move the data at super-high speeds, very low resistance, and very low power. So it's completely unique and that's about one-third of our portfolio are those integrated solutions. We also have clear leadership in eBeam Technology. I talked about our cold field emission electron optics. And that enables us to see those structures.

When you're building Gate-All-Around, and you want to look at the width of those nanosheets, again we have unique technology that enables us to learn faster and then we can co-optimize all of those technologies.

So as we're driving our innovation with customers, we're deeper, we're earlier, we can see fourth generations out, relative to those technologies. So, super, super deep connectivity. That's why also I think Epic is going to be a game-changer. And how we innovate the way we innovate, we talked about the relationship with Leti [ph] for innovation in edge computing and ICAPS. All of those things and then we have our advanced packaging lab in Singapore. That is also a full-flow lab where customers are working on innovation and new architectures.

So, Krish, I think that gives us a tremendous advantage. We can see what's needed earlier. And then the ability to co-optimize all of that gives us a tremendous advantage.

**Q - Krish Sankar {BIO 16151788 <GO>}**

Thanks, Gary.

## Operator

Thank you. One moment for our next question. And our first -- our next question comes from the line of Atif Malik from Citi. Your question, please.

**Q - Atif Malik {BIO 7312618 <GO>}**

All right. Thank you for taking my question. I have a question for Brice. Brice, you talked about China mix normalizing from 45% to 30%. Can you talk about the impact of the mix on the gross margins from the 47.3% you're guiding to for the rest of the year?

**A - Brice Hill {BIO 19639921 <GO>}**

Thanks, Atif. Appreciate the question. So, yes, our gross margin reported in Q1 was 47.9%. We think that -- we've modeled what it would be without the higher China mix. And our view is that the underlying gross margin is approximately 46.7% at this point. So as we go through the course of the year, we expect that gross margin to come down from 47.9% to a more normal amount. At the same time, where we are underneath that, the 46.7% will continue to improve slowly, if that makes sense.

So if you normalized Q1 immediately, you'd be at 46.7%, we expect that to improve through the course of the year. And then we're not changing our goals. Our goals 48% to 48.5% for 2025, that's still what we're targeting as we work on pricing improvements and continue to work on our cost roadmap.

**A - Gary E. Dickerson {BIO 2135669 <GO>}**

Yeah, Atif, I would just add that we have made progress pretty much across all customers on pricing improvements. I think we've talked before about cost headwinds that we encountered in the supply chain. We're making improvements there. And as Brice said, we're committed to hit those goals.

**Q - Atif Malik {BIO 7312618 <GO>}**

Thanks.

## Operator

Thank you. One moment for our next question. And our next question comes from the line of Srini Pajjuri from Raymond James. Your question, please.

### Q - Srini Pajjuri {BIO 5862807 <GO>}

Thank you. Hi, guys. Gary, I have a question on your HBM comment. I think you said HBM accounted for about 5% of industry output. If you can clarify, if that's the wafer output or if that bids [ph]. But my question is, as we look out to the next few years, I think you're forecasting about 50% growth for this business. So it's a little tricky to understand for us, as to how much if DRAM WFE is going to HBM right now. And how do you see that evolving? I mean, if the market grows 50%, should we expect, I guess, equipment spending also to grow 50% or do you think it's going to grow faster than that?

### A - Brice Hill {BIO 19639921 <GO>}

Hi, Srini, since I've seen a lot of the modeling, I'll just share a couple of those data points. So on the first question, it's wafer starts. When we think about the 5% it's wafer starts. I think it is difficult to estimate the equipment purchases at this point because you probably understand that the DRAM business itself has been underloaded, as most of the markets have. So I think what many of the customers are doing is shifting some of their capacity to HBM to get this output.

Gary highlighted in his prepared remarks, that the die sizes for the HBM are larger than the non-HBM. So it certainly will help drive up utilization, which will eventually increase equipment orders going forward. And we do think the DRAM business, if you look at the past few years, the level of WFE for DRAM, we do think that's -- it's been fairly strong and it will continue to be strong, is our expectation. And then the last piece of course is customers are having to expand the HBM-related steps of their DRAM process. Gary highlighted what that is for us and that's growing a lot faster than what you see on the general equipment side.

So my understanding of the DRAM is about 700 steps in the DRAM process and about 15 additional steps. Possibly 20 to do the HBM level of that. So for sure you'll see customers growing the HBM packaging techniques and capabilities, alongside the regular capacity. And then we'll expect to see utilization increase as time goes on.

### Q - Srini Pajjuri {BIO 5862807 <GO>}

Thank you.

## Operator

Thank you. One moment for our next question. And our next question comes from the line of Toshiya Hari from Goldman Sachs. Your question, please.

### Q - Toshiya Hari {BIO 6770302 <GO>}

Hi, thank you so much for taking the question. I had a two-part question. The first one is on the conventional DRAM part of your business and the second part is on your NAND business. So the bullishness in terms of HBM, we understand. I'm hoping to better understand what your customers are doing. What they're telling you, Gary, in terms of their plans on the conventional DRAM side. Are things still very muted and are they

disciplined from a supply perspective or are you seeing a pickup in your conventional DRAM business as well to the extent you have visibility there?

And then I guess a similar question on the NAND side, it's been a soft market for everyone. I guess there's hope that at least node [ph] transitions or layer count increases will resume this year. Are you starting to see early signs of a pickup or are things pretty soft there? Thank you.

### **A - Brice Hill {BIO 19639921 <GO>}**

Yeah, Toshiya. It's Brice. I'll just make a couple of comments and maybe Gary will add to that. So on the DRAM side, and this is true for DRAM and NAND, but I'll start with DRAM. We do see utilization improving, and we also see improvements in prices, and we also see improvements in inventory positions. So we do think that's consistent with the rising optimism on the DRAM side. The utilizations have been low enough that there is a ways to go before they have to start thinking about adding capacity. So our view on the market is it's most -- mostly from a WFE perspective the nodal upgrades and the HBM that we talked about. And it would be similar for NAND, we're seeing improvements in inventory, seeing improvements in pricing, utilization is starting to pick up. And our perspective would be the same as technology advances will be what drives the spending. And we do have signals that as we suggested that the spending will pick up.

### **A - Gary E. Dickerson {BIO 2135669 <GO>}**

Yeah. Toshiya, relative to Applied, in DRAM, as I talked about earlier, we've gained more than 10 points of overall DRAM WFE share over the last 10 years. And then if you look at the technologies for DRAM going forward, periphery moving to higher-speed I/O, enabled by our leadership logic products, capacitor scaling. We're achieving patterning share gains, and I've talked about very strong position in advanced packaging, including high-bandwidth memory. We're really well-positioned there to continue our outperformance in DRAM.

And as Brice said or I said earlier also, we think that business is going to remain very healthy for us. NAND for Applied, we see the revenue up a fair percentage in '24 versus '23, but the total amount is still far below 2022. So that's a little bit more color.

### **Q - Toshiya Hari {BIO 6770302 <GO>}**

Thank you so much.

### **Operator**

Thank you. One moment for our next question. And our next question comes from the line of Harlan Sur from J.P. Morgan. Your question, please.

### **Q - Harlan Sur {BIO 6539622 <GO>}**

Good afternoon, and thanks for taking my question. One of your peers alluded to this on their last earnings call and talked about a pushout on advanced foundry logic programs, due to potentially persistent delays in chip stack [ph] funding. I mean, I think the industry thought that after Congress and the President signed off on that CHIPS Bill. I think it was like 18 months ago that grant funding would be approved -- appropriated at least in 2023. But we are in February '24 and still no grant disbursements. Obviously, all

fab programs will launch at some point, but may be some near-term movements on timing due to the absence of this grant funding. Is that what's driving some of the leading-edge foundry logic programs and delays that you guys talked about in your opening remarks?

**A - Brice Hill {[BIO 19639921 <GO>](#)}**

Okay, Harlan, I'll make a comment there. I do think those scheduled changes that have been in the news. We're up to date on those. So our outlook is consistent with any of those discussions and scheduled changes that you're talking about. On the CHIPS Act, we've recently seen news reports about the government beginning to accelerate that process. We're in that -- we're in the process ourselves of preparing our application on the R&D side and expecting that to open soon. So I think the answer to your question is yes, it is affecting schedules, but we don't expect it will change the ultimate destination of those projects.

**Q - Harlan Sur {[BIO 6539622 <GO>](#)}**

Great. Thank you.

**Operator**

Thank you. One moment for our next question. And our next question comes from the line of Joe Quatrochi from Wells Fargo. Your question, please.

**Q - Joseph Quatrochi {[BIO 18961101 <GO>](#)}**

Yeah. Thanks for taking the question. I was wondering if you could help us and quantify how much your ICAPS business grew in calendar '23. And then as we think about the foundry logic business for 2024, do you think the recovery and leading-edge can offset the declines in ICAPS?

**A - Brice Hill {[BIO 19639921 <GO>](#)}**

Okay, Joe. Thanks. So, I think what we've highlighted publicly on ICAPS is they grew approximately 40% in '22 and it grew faster than that in '23. And so we wouldn't change that and be more specific. But to your point, it's been a strongest market for us, it's now the largest market for Applied. Gary highlighted that there is innovations across that market, it's very important to us from an investment perspective. And so you'll see us continue to focus on serving that market and the growth.

And then the second part was the linearity across the quarters. We're not giving guidance across the quarters, but since we did highlight that we expect some digestion in ICAPS, and we highlighted, we expect leading-edge to accelerate. We'll leave it to you to kind of think about, which is the stronger for us and how the next few quarters go forward. But you know that is the right shape of the -- of those two end-markets.

**A - Gary E. Dickerson {[BIO 2135669 <GO>](#)}**

Yeah, Joe, just let me add. I think our perspective hasn't changed at all relative to how we see the market. So we still see semiconductors at \$1 trillion by 2030. And if you look at, there is some powerful drivers in the digital transformation of every industry. AI is -- certainly there's a lot of focus there and AI server has eight times more foundry logic content, and eight times more DRAM compute memory content.



So as Brice said earlier, and I think as you've heard from others, I think there is a pretty positive perspective on '25. And I think longer-term, we have a very positive perspective relative to semiconductor growth, equipment growing as fast or faster. And Applied outgrowing the equipment market continuing to outgrow as we have for the last five years. So I think quarter-to-quarter or half-to-half, frankly, we don't focus as much on that as we do this secular growth that we see in this industry and the great opportunities Applied has, as I've talked about relative to the major inflections. So anyway, that's the way we think about it.

**Q - Joseph Quatrochi {BIO 18961101 <GO>}**

Got it. Thank you.

**Operator**

Thank you. One moment for our next question. And our next question comes from the line of Timothy Arcuri from UBS. Your question, please.

**Q - Timothy Arcuri {BIO 3824613 <GO>}**

Thanks. I had a clarification and a question. So the clarification, Brice. Is the 2023 WFE baseline, would you agree with something like \$86 billion to \$87 billion? So if you can clarify what your baseline is when you said that you gain share? And then my question is on China WFE. So if I use your numbers, it implies WFE from China is roughly \$30 billion in 2023, maybe a little bit less. I know the customers are not stockpiling tools per se, but we know that SMIC and some of the other public companies, they have revenue to support what they spend. But that seems like only about half of that amount. So I guess the question is, would you disagree with the idea that maybe half of what's coming from China is companies just kind of getting off the ground and trying to displace what's being imported from the U.S. or Europe?

And I guess Gary, the real crux of the question is that the China stuff's not really a free lunch. It's sort of duplicative with spending happening elsewhere. So how do you handicap that when you plan your business going forward? Thanks.

**A - Brice Hill {BIO 19639921 <GO>}**

Okay, Tim. Thank you. So on the 2023 WFE, we've been careful not to engage in the discussion about that. We just shared what our view of Applied's performance and our view of that market. And for us 2023 was a strong year. We talked about ICAPS strength, we talked about DRAM strength, packaging strength, et cetera. And so we'll just have to wait and see what the third parties say about the size of the market for us. We saw a strong market.

On the China WFE, we agree, we don't see stockpiling. There are a number of new customers. So I don't know if it's exactly the partition that you described in terms of leading and public companies versus not. But we do think there are a large number of projects that are under investment where we see over the next four years added wafer start capacity, planned wafer start capacity. That market will be a strong market for us across the planning horizon. And so I think it is a mix. You've got a real demand there. When we look at the macro, we compare the amount of capacity put in place to local China consumption. And we think they're still behind the amount of local China consumption. So we think the investments are rational. And actually, the utilizations look



okay. They're lower than rest of the world, generally speaking. But they're improving and we expect yields to be improving also over time.

And then, I know, Gary is -- you're asking Gary about the free lunch. We think you're right from the perspective of no capacity, we're planning for all of the tools that we sell, whether it's to China or whether it's to a government-incentivized project. None of these things, we think increase the amount of equipment installed, sort of abnormally such that it's not going to be used and not going to serve an end-market. So we don't believe that China demand is an end-line free lunch. We don't believe the government incentives are free lunch from that perspective. It's just affecting the location of needed equipment. Thank you.

**Q - Timothy Arcuri** {[BIO 3824613 <GO>](#)}

Thanks a lot.

**Operator**

Thank you. One moment for our next question. And our next question comes from the line of Joseph Moore from Morgan Stanley. Your question, please.

**Q - Joseph Moore** {[BIO 17644779 <GO>](#)}

Yeah. I wonder if you could give us some clarity on the \$500 million of HBM-related revenue that you're forecasting (Technical Difficulty). And actually, that's a relatively small portion of your overall DRAM run rate at least. And I guess I would have thought it would be even bigger. Can you talk about that? And then there is more than HBM, when it comes to advanced packaging for AI DRAM, people are doing stacks for other types of memory. Are you -- is that HBM kind of encompassing all the advanced packaging or is there other opportunity above and beyond that?

**A - Brice Hill** {[BIO 19639921 <GO>](#)}

Okay. Thanks, Joe. So on the DRAM, I think going back to that Q4, the first quarter we saw elevated DRAM from the China demand. That was the approximation we used for the impact of that. So you're right, it doesn't -- it's not going to exactly describe every single quarter, but I think that's -- that was a good estimate of the incremental that we're seeing. So we'll end up with three straight quarters of incremental DRAM shipping to customers in China for those allowed technologies.

And Gary?

**A - Gary E. Dickerson** {[BIO 2135669 <GO>](#)}

Hey, Joe, on HBM, again the HBM packaging is what we talked about, increasing to almost \$0.5 billion in '24. And our overall packaging -- overall advanced packaging is around \$1.5 billion. So that's kind of how to think about it, about \$0.5 billion in HBM packaging and the total advanced packaging for us is around \$1.5 billion.

**Q - Joseph Moore** {[BIO 17644779 <GO>](#)}

Okay, that's helpful. Thank you.

**Operator**

Thank you. One moment for our next question. And our next question comes from the line of Brian Chin from Stifel. Your question, please.

**Q - Brian Chin {BIO 5763426 <GO>}**

Hi there. Good afternoon, and thanks for letting us ask a question here. Just curious, in route to Applied doing better than the industry over the full year. In terms of the handshake that occurs maybe around midyear, but between some digestion in ICAPS and some pickup in advanced foundry logic. Would the current timing around this suggest maybe a bigger dip in revenue in the July quarter?

**A - Brice Hill {BIO 19639921 <GO>}**

Hi, Brian, it's Brice. So yeah, we're not going to guide future quarters beyond the outlook quarter. So we've given you the shape that we think the end-markets will take. And to your point, it's hard to tell which force will be stronger, whether leading -- growing leading-edge or little digestion on the ICAPS side. So we're not going to call that until we get to those quarters.

**Q - Brian Chin {BIO 5763426 <GO>}**

Okay. Fair enough. Thanks.

**Operator**

Thank you. One moment for our next question. And our next question comes from the line of Charles Shi from Needham & Company. Your question, please.

**Q - Charles Shi {BIO 22238518 <GO>}**

Hi. Thanks for taking my question. I have a long-term question for leading-edge foundry logic. So you guys talk a lot about the material engineering potentially driving outperformance. If I look at the back, in the last 10 years, I think at the back half of the last decade, I mean, between '15 and then 2019. Because the manufacturers kind of slow to jump on EUV, there was a lot more adoption of multi-patterning, actually led to like outperformance of the (inaudible) and edge, equipment suppliers like Applied Materials. But the last five years, because of the EUV adoption, seems to be, that trend has reversed a little bit. But looking out for the next five years, I know, there are recent discussions about maybe high-end EUV may not actually get adopted before 2030. Does that -- does Applied think about mainly to more of the multi-patterning EUV again, and that could actually drive up the material engineering intensity again. And any thoughts would be helpful. Thank you.

**A - Brice Hill {BIO 19639921 <GO>}**

Yeah, thanks for the question. So one thing I would point to -- one of our largest customers, they talked about what's driving their roadmap going forward. They talked about something called design technology co-optimization. So what they said basically was that much of the area scaling they were driving going forward is coming from new structures and new materials. So an example is backside power. You can get 30% area savings through that type of a structure with no change in feature size.

So I think certainly what we see, and we're working with customers for technology nodes out, past the end of the decade we see the relative contribution of materials innovations,

spending going higher, the percentage of that going higher. Gate-All-Around, backside power, there's the CFAT [ph] technology, there is many different innovations, packaging technologies. All of those areas we have over 50% share opportunity in those inflections that are very accretive. And again, we do see the relative contribution from -- of spending for those innovations to go higher over time.

I'll give you one more data point. So Gate-All-Around, Gate-All-Around is a new innovation in the transistor to process the data faster. And we see Gate-All-Around ramping to more than \$1.5 billion for Applied revenue in '24, and almost double that amount in calendar '25. So again those are very powerful new architecture inflections where Applied is extremely well positioned.

**A - Michael Sullivan {[BIO 16341622 <GO>](#)}**

Okay. Thanks, Charles. And operator, we're getting close to the end of the session. So if we have time for one more quick question, please.

**Operator**

Certainly. One moment for our final question for today then. And our final question for today comes from the line of Thomas O'Malley from Barclays. Your question, please.

**Q - Thomas O'Malley**

Hey guys, thanks for sneaking me in. I had another question on kind of the handoff from the first half to the second half. Clearly you're kind of talking about the ICAPS business, getting a little softer in the back half but leading edge is really picking up slightly offsetting. In terms of where you're seeing the strength in the leading edges. Is that greenfield, new fab buildouts, or is that existing capacity additions? Any kind of help on where that strength is coming from in the second half would be helpful. Thank you.

**A - Brice Hill {[BIO 19639921 <GO>](#)}**

Sure, Tom. It's Brice. Typically it's greenfield. So I think that when companies start the first part of a process, you're typically putting in greenfield and you will shift some of your reuse equipment later if you're able to do that. So that would be my expectation. Thank you.

**Q - Thomas O'Malley**

Thank you.

**A - Gary E. Dickerson {[BIO 2135669 <GO>](#)}**

Thank you.

**A - Michael Sullivan {[BIO 16341622 <GO>](#)}**

Okay. Thanks, Tom. And I appreciate that question. Brice, how would you like to give us your closing thoughts for today?

**A - Brice Hill {[BIO 19639921 <GO>](#)}**

Sure, Mike. What stands out to me from a summary perspective is that we've anticipated the major market trends and we work closely with our customers to invest in the most important technology inflections. I think we'll be a major beneficiary as AI and IoT spending grows over the next several years. Our number one positions in Gate-All-Around, backside power, and advanced packaging are higher than our corporate average, which gives me confidence that we'll continue to gain share. Beyond our strong portfolio, we're also making operational progress, which makes me confident we can meet strong demand and make progress in gross margins.

Finally, our services growth is accelerating to double-digits and generating more than enough profit to fund our growing dividend. Also, I hope to see many of you at the Morgan Stanley Conference on March 4th. Mike, thank you. Let's close the call.

## A - Michael Sullivan {[BIO 16341622 <GO>](#)}

Okay. Thanks, Brice. And we'd like to thank everybody for joining us today. A replay of today's call is going to be available on the IR page of our website, by 5:00 Pacific Time. And we'd really like to thank you for your continued interest in Applied Materials.

## Operator

Thank you, ladies and gentlemen, for your participation in today's conference. This does conclude the program. You may now disconnect. Good day.

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